

Hierarchical Multi-Agent Reinforcement Learning for Autonomous Cyber Defense in Coalition Networks

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Abstract—This paper presents a comprehensive methodology for autonomous cyber defense (ACD) through the deployment of distributed agents using multi-agent reinforcement learning (MARL). The primary objective is to maintain operational services distributed across three geographical locations interconnected by an untrustworthy network infrastructure (Internet). The ACD agents will be developed and tested in the Cyber Operations Research Gym (CybORG) environment, which simulates a wide range of network conditions and hosts both red (attacker) and blue (defender) agents. The proposed solution employs a centralized training approach with a decentralized execution strategy using Multi-Agent Proximal Policy Optimization (MAPPO). In addition, we discuss the use of an efficient communication protocol to digitize messages into an 8-bit format, using personalized message generation for the ACD agents, thereby enhancing their collaborative defensive capabilities. Furthermore, we outline a curriculum learning methodology and a hierarchical agent structure to optimize defense strategies and zone-specific tasks.

Index Terms—Autonomous Cyber Defence, Multi-Agent Reinforcement Learning, Cyber Security

I. INTRODUCTION

The cyber threat landscape is undergoing a period of rapid evolution, driven by the convergence of human ingenuity and recent advances in generative Artificial Intelligence (AI). For example, recent research has leveraged Large Language Models (LLMs) to automate cyber attacks with human-level performance [1]–[3]. To counter these sophisticated threats, AI models are also being used to implement and enforce countermeasures. In particular, Multi Agent Reinforcement Learning (MARL) agents are able to learn and adapt to new attack strategies, making this approach more robust than static defenses and reducing the reliance on human intervention. This approach distributes the learning process across multiple agents, allowing it to scale to large and complex networks segmented into different security zones.

In this paper, we explore methods to demonstrate that MARL agents can effectively solve complex cyber defense tasks. We discuss the use of curriculum learning to progressively train the agents, starting with simpler scenarios and gradually increasing complexity. We also design a hierarchical agent structure, with high-level agents coordinating overall defense strategies and low-level agents focusing on zone-specific tasks. In addition, we implement a context-aware 8-bit communication protocol to facilitate efficient inter-agent communication despite bandwidth constraints. Together, these methods enable the MARL agents to dynamically and robustly respond to evolving cyber threats.

This research builds on the network scenario from the Cyber Autonomy Gym for Experimentation (CAGE) Challenge 4 [4], simulated with Cyber Operations Research Gym (CybORG) [5], [6]. A total of five agents are deployed across three different security zones, with each zone being connected to the others by an untrustworthy network infrastructure. The goal of each agent is to protect their respective zone against red agents, which may access the untrusted network via a contractor network. Communication between Autonomous Cyber Defense (ACD) agents is constrained by link data rate, message size, and security zones (e.g., one-way links between operational and restricted network zones). As a result, compiling a centralized view of the entire network is challenging, requiring the decision-making process to rely on a mixture of local observations and shared messages of restricted size. These limitations motivate the use of Centralized Training and Decentralized Execution (CTDE) to achieve satisfactory performance within the constraints of the environment.

The main contributions of this paper are as follows:

- 1) The design of a multi-agent solution using CTDE and a context-aware 8-bit communication protocol between distributed agents.

- 2) A methodology for curriculum learning of MARL defense agents, incorporating various training scenarios.
- 3) The definition of a hierarchy of specialized blue agents performing high and low level tasks in restricted and operational zones.

The remainder of this paper is organized as follows. Section II discusses related work that also uses MARL for autonomous cyber defense and the motivation for the present investigation. Section III defines the network scenario and the problem addressed in this research. Section IV describes the autonomous agent architecture and discusses the main functional blocks in detail. Section V discusses the difficulties encountered while trying to achieve the previously defined goals, as well as possible causes and solutions. Finally, section VI concludes the paper and lists future work.

II. RELATED WORKS

Recent research proposes multi-agent strategies to optimize joint policies for improved cooperation among agents. For example, the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm [7] addresses the non-stationary nature and high variance in MARL through a combination of centralized training and decentralized execution. Similarly, Lohse et al. [8] proposed an extension for the Proximal Policy Optimization (PPO) algorithm [9] that applies the clipped surrogate objective function to a centralized critic in the setting of job shop scheduling problems. Centralized training stabilizes the training by accessing the actions and observations of all agents, while actors use only local observations, which improves the consistency and robustness of the training and allows execution with local information.

The MARL concept has also been adapted for hierarchical agent teams for autonomous cyber defense and automated penetration testing [10], [11]. For example, Yunlong et al. [11] propose a hierarchical MARL network attack and defense game, as well as a cooperative defense decision-making method that autonomously and efficiently completes the formulation of defense strategies and responses. In addition, Abegaz et al. [12] introduced an intelligent resource trading framework that incorporates MARL, blockchain technology and game theory to regulate dynamic resource trading environments, with security and confidentiality enforced by smart contracts in a consortium blockchain. Similarly, Chen et al. [13] proposed a decentralized secondary control approach using the Asynchronous Advantage Actor Critic (A3C) algorithm for MARL, combining centralized offline learning for overall system optimization with decentralized online execution. They also introduced a proactive event-triggered communication mechanism to mitigate Denial of Service (DoS) attacks and reduce communication resource consumption.

Another major challenge is inter-agent communication in low-bandwidth network scenarios. Li et al. [14] proposed Context-Aware Communications (CACOM), a communication protocol designed to improve the efficiency of MARL under

communication constraints. CACOM employs a two-stage communication process in which agents broadcast coarse context messages for situational awareness, followed by personalized messages using attention mechanisms to ensure relevance.

In penetration testing, Ghanem et al. [10] propose an approach called the Intelligent Automated Penetration Testing Framework (IAPTF), which uses model-based Reinforcement Learning (RL) to automate sequential decision making. Penetration testing tasks are treated as a Partially Observed Markov Decision Process (POMDP), which is solved with an external POMDP solver using various algorithms to identify the most efficient options.

Despite these advances, there remains a research gap in the context of automated cyber defense in effectively coordinating multi-agent defense strategies in highly segmented networks with limited communication bandwidth. This research addresses this gap by discussing a hierarchical approach using Multi-Agent PPO (MAPPO) and a context-aware communication protocol to enhance collaborative defense capabilities in ACD scenarios.

III. NETWORK SCENARIO & PROBLEM

This research utilizes the network scenario from CAGE Challenge 4 in [4], which consists of four interconnected networks using an untrusted network infrastructure, as shown in Fig. 1. The number of user hosts is randomized between 3-10 while the number of servers is randomized between 1-6 in each security zone, with up to 5 services per node. The networks A and B have two security zones, restricted (A_1 and B_1) and operational (A_2 and B_2). Each zone hosts an ACD blue agent. Meanwhile network C has one ACD blue agent that is responsible for three zones, namely the Admin- and Office Network, as well as the DMZ/Public Access Zone. As a result, the agents are distributed across the network and cooperate using messages of restricted size and limited link data rate. These limitations force the agents to make decisions mainly using local information with limited access to the shared knowledge, since it is not feasible to compile a comprehensive centralized view of the network state within the communication constraints. The undefended contractor network (D) hosts the red agents that attack the network using the unsafe infrastructure. There are three ways to initiate a cyberattack: the red team selects a computer in the contractor network and launches a cyberattack via the untrusted network; a user (green agent) opens a phishing email; or a user accesses a compromised service that triggers a cyberattack.

The primary task of the ACD agents that make up the blue team is to ensure operational capabilities while preventing malicious activity on the network to maintain a stable cyber posture. If possible, enterprise servers should be maintained to ensure that less sensitive day-to-day operations are not disrupted and to provide access to public services provided by headquarters. This can be difficult to uphold because operational priorities may change over time, requiring

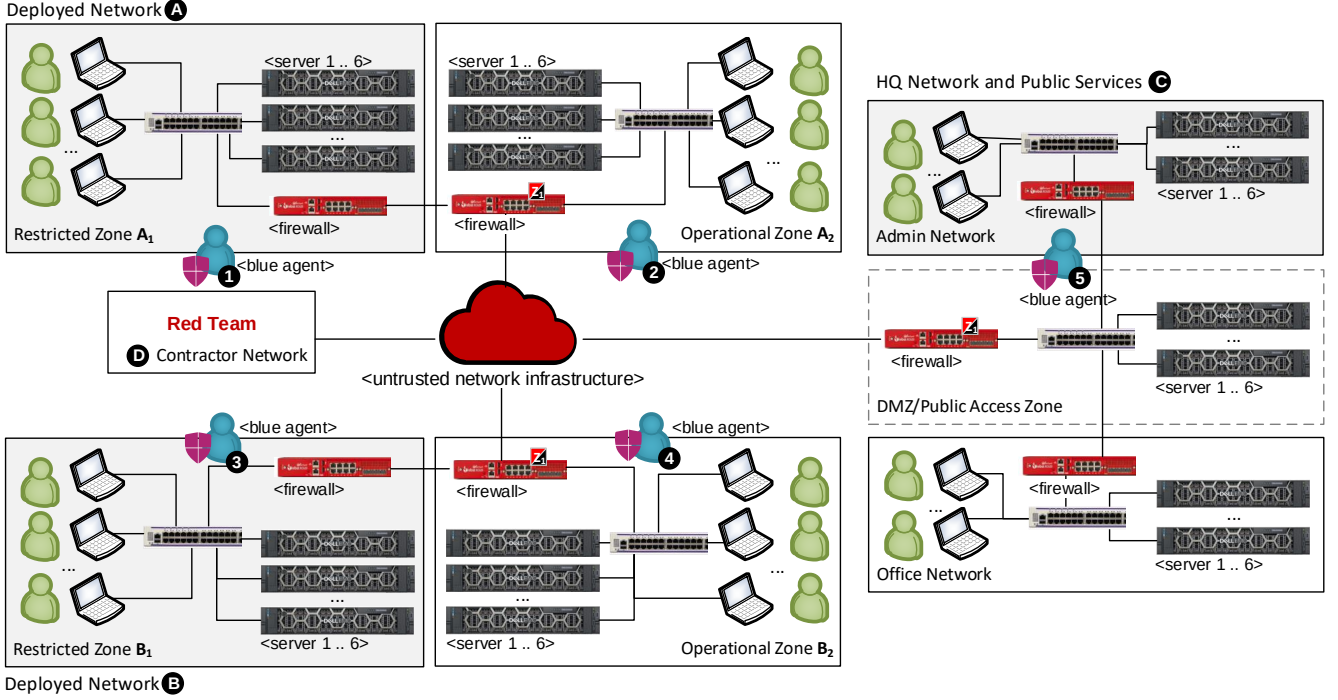


Fig. 1: Scenario with four networks (A-D). Deployed networks A and B each host two blue agents, while HQ network C hosts a fifth. Red agents hosted on contractor network D can use the interconnecting untrusted network infrastructure as entry point.

downtime of operations and services depending on different phases or objectives of the mission.

The agents operate in three phases with distinct priorities and network connectivity rules as defined by the challenge [4]:

- Phase 1: General low-priority operations in all zones
- Phase 2a: Increased activity and focus in specific zones
- Phase 2b: Transition of focus to other critical zones

This results in blue agents having significantly different priorities regarding their actions depending on the phase.

Users are represented by green agents, which are present on every host. Green agents randomly perform local work or attempt to reach external services within their local or remote zones. The service and zone a green agent chooses are randomly determined but limited to valid connections for the current mission phase. Green agents occasionally generate false alarms while performing their tasks by exhibiting behavior similar to red agents, such as transferring data between hosts. They also have a low probability of introducing red agents into the network by succumbing to phishing attacks, installing unauthorized software, and practicing poor security hygiene.

A. Problem

The problem faced by the blue agents is defined as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) $M = \langle S, A, P, R, \Omega, O, n, \gamma \rangle$. It involves maintaining operational capabilities while preventing malicious activity on a segmented network that is highly dynamic and contains different levels of security zones. In this context, n

represents the number of blue agents defending the network, and $\gamma \in (0, 1)$ is the discount factor used to balance immediate and future rewards. At each time step t , the environment is in some state $s \in S$. Each blue agent i receives a local observation $o_i \in \Omega$ drawn from the observation function $O(s, i)$. This local observation contains information about the hosts for which the agent is responsible, including potential alerts triggered by green or red agents.

Agents communicate with each other within the constraints of limited bandwidth and restricted communication channels to exchange local information. After this communication phase, each agent follows its local policy π_i to select an action $a_i \in A$ based on its observations and received information. These individual actions form a joint action $\mathbf{a} \in A^n$, which causes the environment to transition to a new state $s' \sim P(s' | s, \mathbf{a})$ and generates a global reward $r = R(s, \mathbf{a})$.

The main goal for the blue agents is to maximize the expected sum of discounted rewards, $G_t = \sum_t \gamma^t r_t$. Rewards are built to penalize failures to maintain service availability and detect red agent activity, while rewarding successful defense actions and maintenance of critical services. Each agent maintains a local history of actions and observations to inform future decisions and adapts its strategy as it learns more about the state of the network and the behavior of the red agents.

IV. METHODOLOGY

A. Centralized Training and Decentralized Execution (CTDE)

This investigation starts at our recent research [1] and the hypothesis that it is possible to train a multi-agent system to defend the network zones defined earlier in section III. Each agent manages a specific network zone and must adapt its defense strategies based on network changes and threat evolution. Fig. 2 shows an abstract representation of the gym environment that hosts the multi-agent architecture. In this figure, the centralized critic is used only during training (A) while at execution (B) the agents are distributed in the network using the CTDE paradigm.

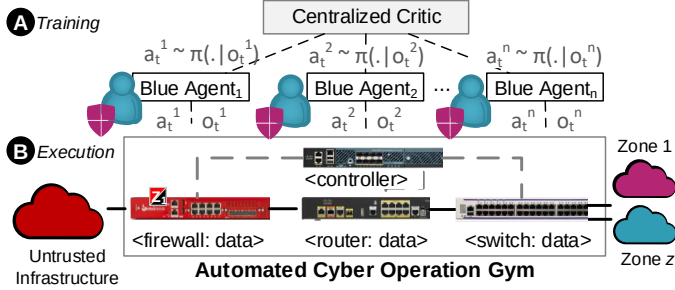


Fig. 2: Multi-Agent architecture using CTDE.

The rewards for the blue agents are calculated according to their ability to maintain the operational integrity of the network and neutralize the activities of the red agents. The reward structure reflects the criticality of various hosts and their roles within the various mission phases.

The initial score for blue agents is zero. Penalties are assessed when green agents, representing regular users, are unable to perform their assigned tasks due to an inability to connect to the required services or the unavailability of their host. Penalties also occur when green agents gain access to services that have been compromised by red agents, or when red agents successfully perform disruptive actions such as the *Impact* action.

Penalties also depend on the specific mission phase and the importance of the affected zones. During the pre-mission planning phase (Phase 1), the penalties are relatively evenly distributed across the different zones. However, during the mission-specific phases (Phase 2a and Phase 2b), the penalties for failing to maintain services in operational zones increase significantly, reflecting their increased importance.

B. Hierarchy of Autonomous Cyber Defence Agents

To optimize the defense of a segmented network, a hierarchical structure of agents is recommended, each responsible for a specific security zone and its associated defense requirements. High-level agents oversee multiple security zones and develop and execute strategies to ensure network integrity and availability. They adjust firewall policies based on the mission phase and allocate resources as needed to maintain critical communications during high-priority phases and protect against coordinated attacks.

Low-level agents are responsible for local defense within individual security zones. They have specialized knowledge tailored to the specific needs and vulnerabilities of their respective zones. In zones that are restricted and contain sensitive data, low-level agents deploy decoy services, perform malware analysis, and quickly remediate compromised systems. In operational zones, their primary goal is to maintain service availability and mitigate detected threats to minimize disruption.

Two examples of high-level agents are the Central Command Agent (CC-A) and the Strategic Analysis Agent (SA-A). The CC-A oversees the comprehensive network defense strategy, which includes isolating zones to prevent threat propagation, maintaining critical communications pathways, and deploying decoy services to mislead potential attackers. The SA-A is responsible for aggregating and analyzing information across zones, monitoring flagged malicious activity, investigating suspected malware, and initiating system remediation.

In contrast, low-level agents in restricted zones are primarily concerned with continuous monitoring, threat remediation, and access control enforcement. The Monitor Agent (M-A) scans for malicious activity, deploys deception services, and collects threat intelligence. The Remediation Agent (R-A) is responsible for removing threats and restoring systems. The Access Control Agent (AC-A) is responsible for managing traffic to prevent unauthorized access.

In operational areas, low-level agents are responsible for ensuring service continuity, responding to incidents, and managing network traffic. The Service Continuity Agent (SC-A) is responsible for monitoring disruptions, restoring compromised services, and providing decoy services. The Incident Response Agent (IR-A) is responsible for remediating threats, investigating alerts, and scanning for new incidents. The Traffic Management Agent (TM-A) is tasked with blocking malicious traffic, ensuring essential communication paths remain open, and monitoring traffic patterns for anomalies. This hierarchical approach ensures robust and adaptive defense across the network.

C. Communication between Blue Agents

In the context of our scenario, with a network segmentation into multiple security zones and agents operating under partial observability, effective inter-agent communication is critical for achieving high rewards. Each agent, responsible for specific zones, needs to share local observations and derived insights to improve the collective defense strategy. We consider a shared message space $\mathcal{M}$, where a message $m \in \mathcal{M}$ can be a scalar or a vector. The communication policy of agent $i \in \mathcal{D}$, denoted ξ_i , generates messages based on the agent's observation or interaction history. At each step t , agent i sends a communication message $m_{i \rightarrow j}^{(t)}$ to agent j for all $j \neq i$. This communication policy allows agents to effectively coordinate their actions, ensuring a synchronized defense across multiple zones. Given the limited bandwidth and the need for efficient communication in our network, we digitize messages to 8 bits

in order to overcome these constraints and make the message computation learnable with respect to the original loss of the MARL approach.

In our scenario, network connectivity and communication policies are defined for different mission phases. Each mission phase has an associated communication policy that governs accessibility between zones. When the mission phase changes, the intended policy is automatically communicated, affecting only the connections associated with that mission. For example, when Mission 2A is enabled, only connections with Restricted Zone A and Operational Zone A are affected. The intended policy and actual firewall state are also communicated to blue agents in their observation vector. Blue agents can open and close firewalls between their zones and other networks to prevent infections from red agents, but they may be penalized if their changes prevent green agents from achieving their respective objectives.

Blue agents have the ability to communicate 8-bit messages with each other, regardless of firewall policy. This is critical for maintaining coordination across security zones. During the pre-planning phase (Phase 1), the network connectivity policy allows broad communication between most zones, facilitating initial coordination and planning. However, during the active mission phases (Phases 2A and 2B), connectivity is more restricted to ensure that critical operations are protected from potential breaches. For example, during Phase 2A, Operational Zone A is inaccessible from all other networks except the HQ, emphasizing the importance of maintaining high-priority communications while protecting other zones.

D. Curriculum Learning

The curriculum learning approach is designed to incrementally train blue agents to handle increasingly complex and realistic scenarios. This method ensures that agents develop a solid understanding of the basic principles before tackling more sophisticated challenges. Training begins with simpler scenarios involving fewer hosts and red agents. More complex scenarios with increased randomization in network configuration and red agent strategies are gradually introduced. For example, the list below defines six network scenarios of increasing complexity.

1) *Single Zone, Minimal Host Scenario*: A single network zone hosting 1-2 servers and 3-5 user hosts, and an active red agent with limited capabilities. The goal is to maintain service availability and detect the red agent.

2) *Two Zones, Limited Interaction Scenario*: A deployed network with both restricted and operational zones, each with 2-3 servers and 4-6 user hosts. Two red agents with basic attack strategies. The goal is to secure both zones while managing communication between them.

3) *Basic Deception Scenario*: A deployed network with both restricted and operational zones, each with 3-4 servers and 5-7 user hosts. Possible deployment of decoy services to mislead attackers. Red agents are implemented with moderate

attack strategies. The goal is to use decoys effectively while maintaining overall network security.

4) *Multiple Network Scenario*: Both deployed networks are active with their respective restricted and operational zones. Each zone has 4-5 servers and 6-8 user hosts. Four red agents with advanced attack strategies are deployed. The goal is to coordinate defense across multiple zones and manage complex inter-zone communications.

5) *Full Network, High Randomization Scenario*: All networks (HQ, deployed networks A and B, and contractor network) are active. Random number of hosts and services within each zone (1-6 servers, 3-10 user hosts). Red agents use highly randomized and sophisticated attack strategies. The goal is to maintain operational integrity in a fully active network with high variability.

6) *Dynamic Mission Phase Scenario*: Full network with three dynamic mission phases that affect zone priorities, namely Phase 1: *General*, Phase 2A: *Deployed Network A Priority*, Phase 2B: *Deployed Network B Priority*. Adaptive Red Agent strategies based on mission phase. The goal is to adapt defense strategies in real time to changing mission priorities and red agent tactics.

Given these six scenarios, the training strategy can be structured into the following phases.

a) *Initial training phase*: Start with Scenario 1 to allow agents to learn basic defense mechanisms. Focus on maintaining service availability and detecting simple red agent strategies. Use rewards and penalties to reinforce correct behavior and punish mistakes.

b) *Progressive complexity*: Gradually introduce scenarios 2 and 3, increasing the number of zones and complexity of red agent strategies. Include decoy services and basic inter-zone communication. Continually monitor agent performance and adjust training parameters as needed.

c) *Advanced training phase*: Proceed to scenarios 4 and 5, activating multiple networks with high randomization in network configurations and red agent tactics. Train agents to coordinate across multiple zones and handle more sophisticated attacks. Implement curriculum learning principles by incrementally increasing scenario complexity.

d) *Real-Time adaptation*: Introduce Scenario 6, where agents must adapt to dynamic mission phases and changing priorities. Train agents to adjust their strategies in real time based on the current mission phase and red agent behavior. Use advanced RL techniques such as hierarchical coordination and prioritized experience replay to increase learning efficiency.

As a result, this methodology enables continuous evaluation of agent performance through rigorous testing in increasingly complex scenarios. Metrics such as service availability, threat detection rate, and response time are used to evaluate performance and effectiveness. Agent robustness is the result of iterating on training strategies based on evaluation results, refining agent behavior and improving overall performance.

V. DISCUSSION

In the course of our ongoing research, we encountered a number of challenges that resulted in suboptimal performance of the trained blue team. Initial implementations using vanilla models with CTDE approaches such as MAPPO or Deep Deterministic Policy Gradient (DDPG) did not show significant learning progress in complex environments. These environments included dynamic network setups with varying numbers of hosts per subnet and the three distinct phases of the scenario without the use of curriculum learning.

The complexity of the problem arose from the separation of the episodes into three phases, each of which significantly changed the optimization objective. In MARL settings, the dynamic nature of the autonomous agents causes the environment to be non-static from the perspective of a single agent, violating the Markov property that guarantees convergence for RL methods. CTDE approaches such as MAPPO attempt to mitigate this problem by using a centralized critic that computes action values based on the global state and the actions of all agents. However, the additional dynamic feature of fluctuating goals in each phase in the CybORG environment complicates these efforts. To address this, we trained the blue agent suite on isolated subsets of the environment, first optimizing performance on specific phases and then gradually fine-tuning the models for subsequent phases. This approach is akin to training on randomly selected sub-policies for each episode to achieve robustness to policy changes in competing agents [7]. In our case, this involved adapting to the changing phases of the environment that required different policies or action priorities. Implementing a simple curriculum learning approach, that trains agents sequentially on different phases for 500,000 steps each, helped generating effective policies for the distinct phases. This insight is further supported by the analysis of Shapley values which revealed that the actions chosen by each agent are highly dependent on the phase and the connected subnets, further highlighting the need for phase-specific training to develop robust policies.

Currently, we are implementing a learnable message computation system that encodes binary subnet information and agent observations for transmission over low-bandwidth links. However, the information exchanged is not specialized for the receiving agent. A context-aware implementation using Learned Step Size Quantization (LSQ) and transformer attention mechanisms could improve this approach by generating personalized messages.

VI. CONCLUSION

This paper describes an ongoing investigation to train and test a team of blue agents distributed in a dynamic network scenario using MARL to learn near-optimal policies within an action space. In addition, we intend to improve our training schedule to include different red agent strategies, curriculum learning with different environments, and the aforementioned training on subsets of the actual episodes that include only

certain problem phases before fine-tuning on subsequent phases. For example, recent research uses randomly selected subsets of each episode to obtain agents that are robust to changes in the behavior of competing agents. This could be applied to our scenario in two ways: (I) building resilience to strategies employed by more powerful red agents, as well as (II) the dynamics of different episode phases with unique optimization goals. Finally, we intend to explore different techniques for message quantization, possibly including LSQ and context-aware messages using attention mechanisms.

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